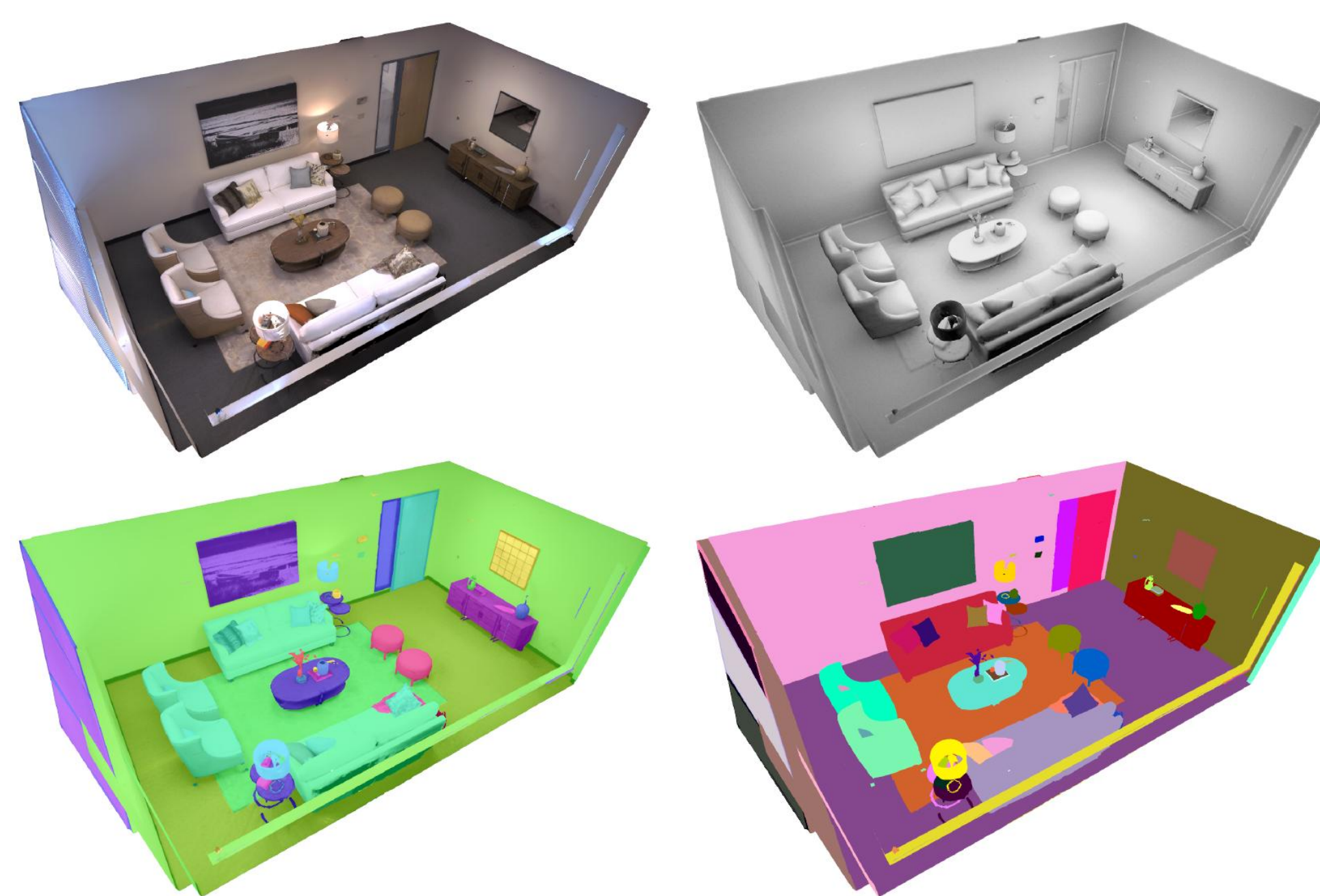


Data and Pre-Processing

Replica Dataset

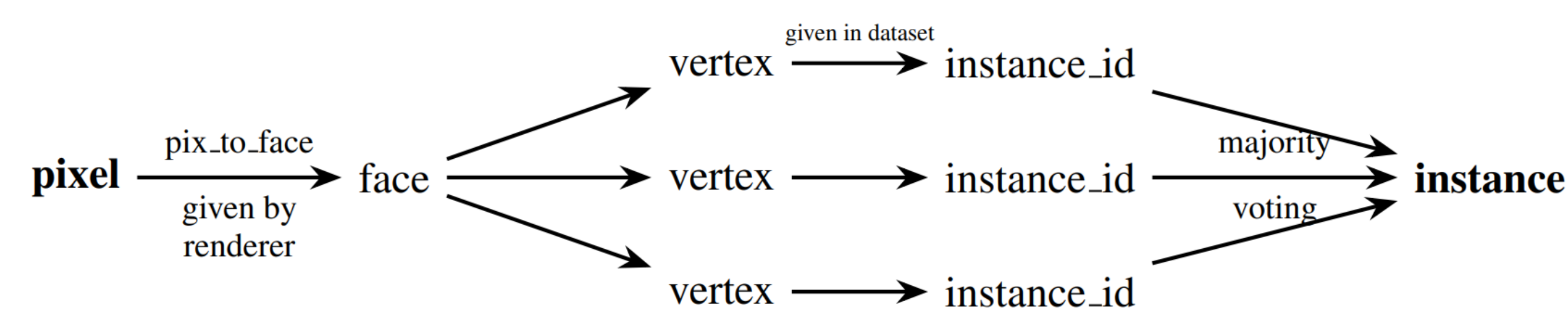
High quality 3D mesh indoor data

- 6 training scenes: 4 offices, 2 rooms
- 2 evaluation scenes: 1 office and 1 room



Generating Instance Masks

Render 2D views from 3D scenes, then get instance masks by voting, using vertex annotations from Replica



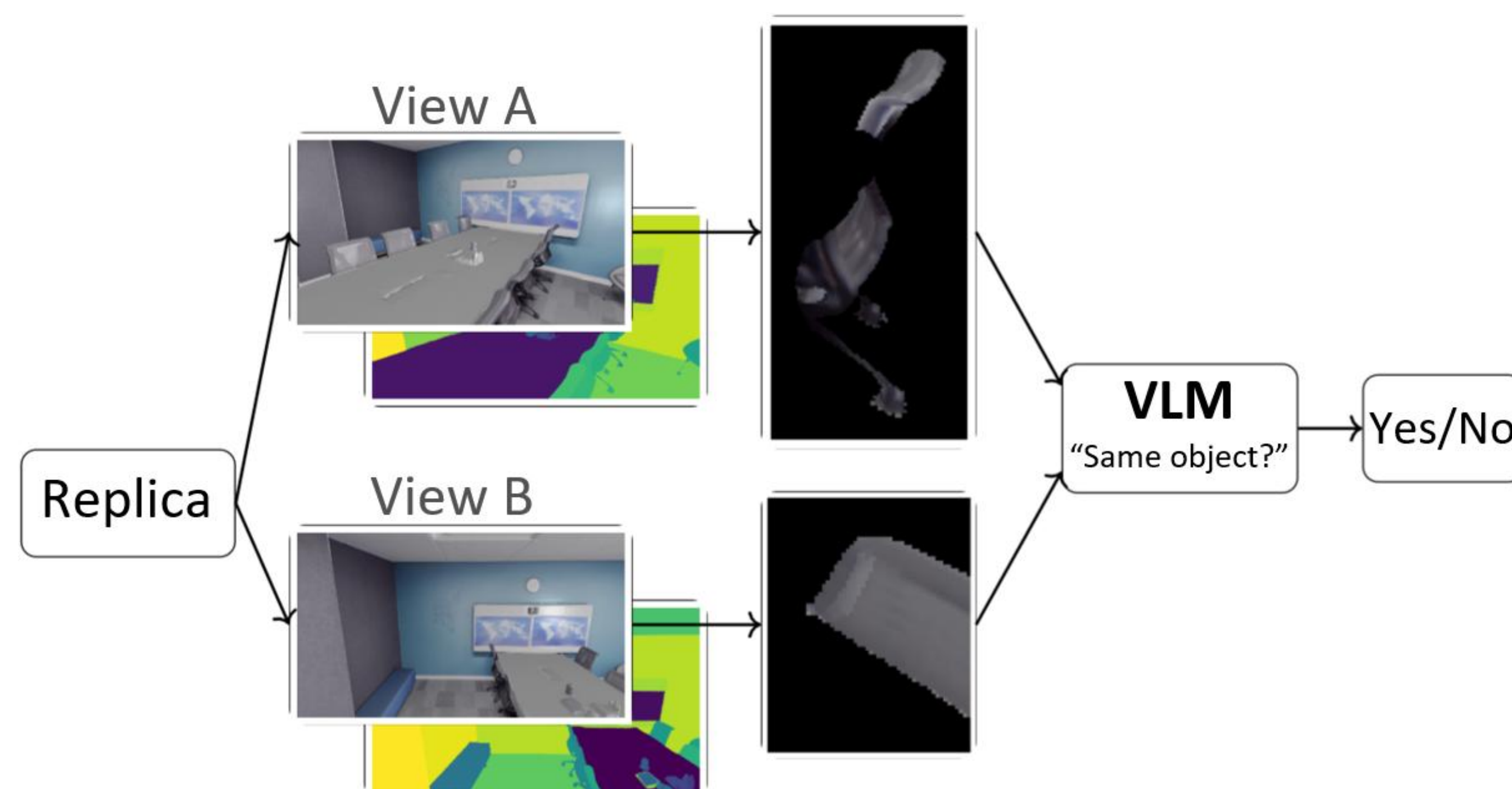
Constructing Instance Pairs

- Sample two views randomly
- Shared instance IDs defines positive pair, else the pair is negative
- Crop around instance and add padding

Methods

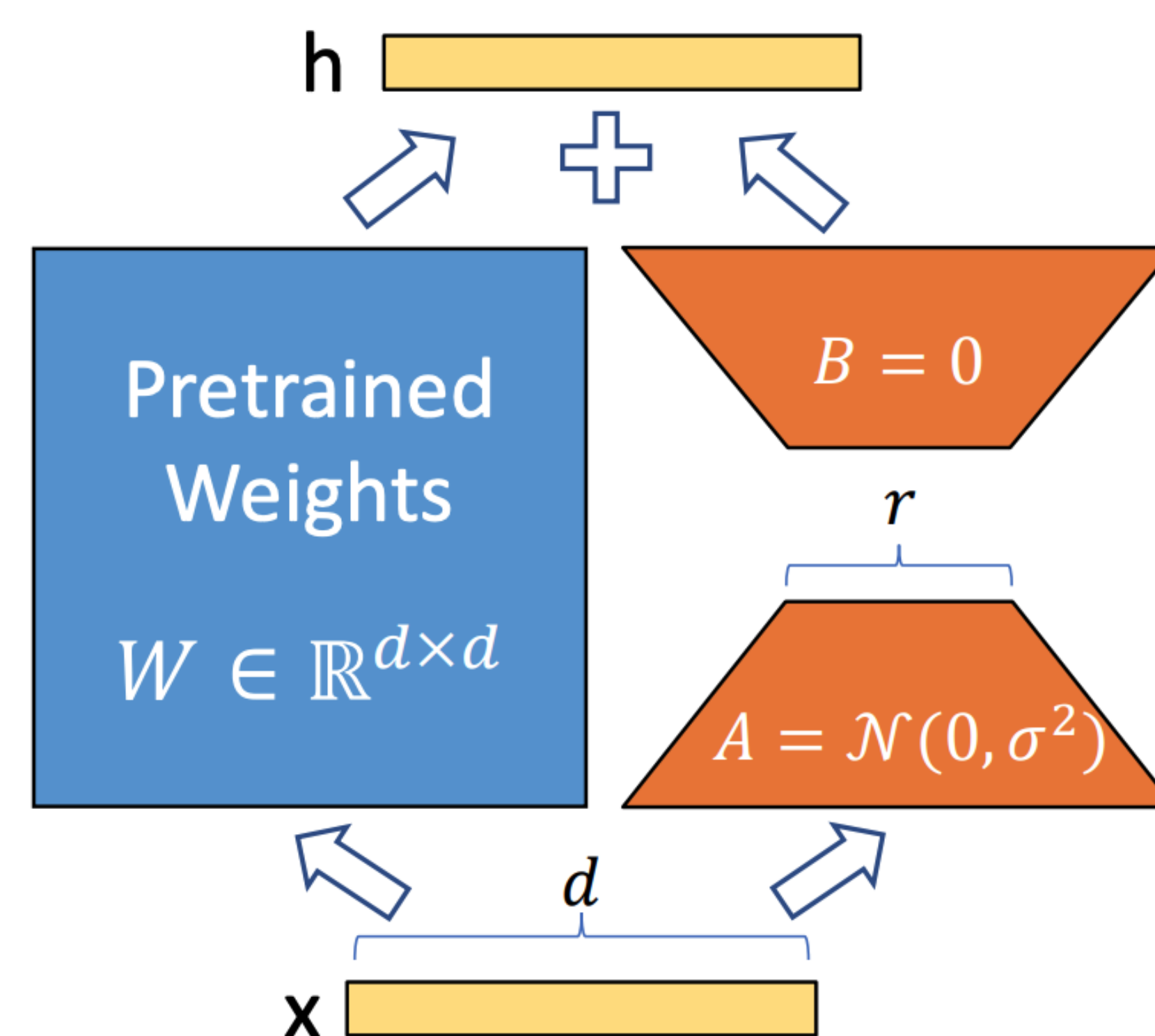
Using VLMs to group 2D views

- Pairs of instance crops are fed to the VLM
- We ask if they are views of the same object
- A base version of Qwen3-VL is compared to a LoRA-fine-tuned version



LoRA Fine-Tuning

- Freeze original model weights
- Inject low-rank matrix updates
- $h = Wx + BAx * \alpha/r$



Results

Performance Comparison

- Fine-tuning improves accuracy and precision significantly, suggesting it helps internalize task-specific cues
- Qwen consistently outputs yes or no answers

Performance of Base vs Fine-Tuned model

Scene	Model	Acc.	Prec.	Rec.	Amb.
office4	Base	0.74	0.67	0.88	0.00
office4	LoRA	0.82	0.76	0.92	0.00
room2	Base	0.80	0.76	0.91	0.00
room2	LoRA	0.92	0.91	0.94	0.00

Failure Modes

Typical failures include:

- Occlusion from other objects
- Symmetry and repetitiveness
- Very small crops

Example of positive pair correctly classified by the fine-tuned model but not by the base model. Probable failure mode: occlusion and symmetry.

